

Trainings ▾

General Information

Cummins Inc. recommends the use of fully formulated antifreeze or coolant containing a precharge of Supplemental Coolant Additive (SCA). The antifreeze or coolant **must** meet the specifications outlined in The Maintenance Council (TMC) Recommended Practice (RP) 329 (ethylene glycol) or RP 330 (propylene glycol). The use of fully formulated antifreeze or coolant significantly simplifies cooling system maintenance.

Copies of TMC specifications can be obtained through Cummins Inc., or by contacting:

The Maintenance Council
American Trucking Association
2200 Mill Road
Alexandria, VA 22314-5388
Phone: (703) 833-1763
Fax: (703) 836-6070

Fully formulated antifreeze contains balanced amounts of antifreeze, SCA, and buffering compounds, but does **not** contain 50 percent water. Fully formulated coolant contains balanced amounts of antifreeze, SCA, and buffering compounds already premixed 50/50 with deionized water.

The following pages give an explanation of water, antifreeze, and SCA's. They also explain how to test antifreeze and SCA levels.

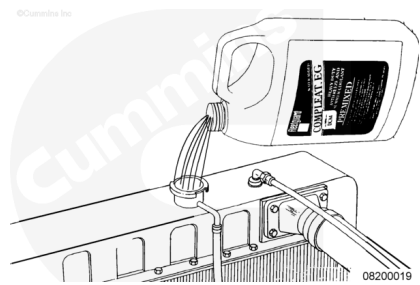
This section also contains information on cooling system maintenance and a coolant treatment chart that is used to determine the correct SCA service filter.

Alternative maintenance practices for cooling systems can be found in Cummins Coolant Requirements and Maintenance, Bulletin 3666132 (</qs3/pubsys2/xml/en/bulletin/3666132.html>).

Fully Formulated Coolant/Antifreeze

Cummins Inc. recommends using either a 50/50 mixture of high quality water and fully formulated antifreeze or fully formulated coolant when filling the cooling system. The fully

formulated antifreeze or coolant **must** meet TMC RP 329 or TMC RP 330 specifications.



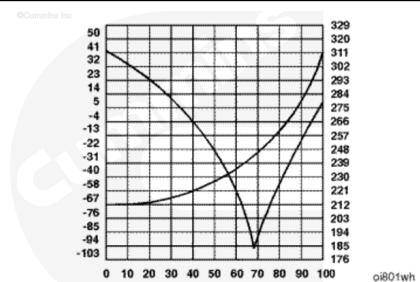
High quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfur	100 ppm as (SO ₄)

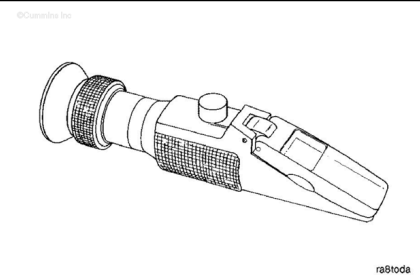
Cummins Inc. recommends using Fleetguard® Compleat. It is available in both glycol forms (ethylene and propylene) and complies with TMC standards.



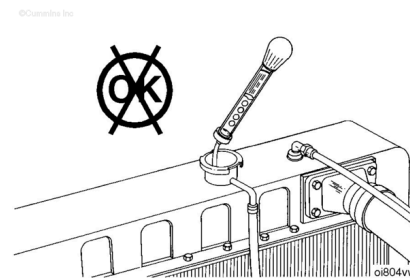
Fully formulated antifreeze **must** be mixed with high quality water at a 50/50 ratio (40 to 60 percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-34°F] freezing point and a boiling point of 110°C [228°F], which is adequate for locations in North America. The actual lowest freezing point of ethylene glycol antifreeze is at 68 percent. Using higher concentrations of antifreeze will raise the freezing point of the solution and increase the possibility of a silica gel problem.



A refractometer **must** be used to accurately measure the freezing point of the coolant.



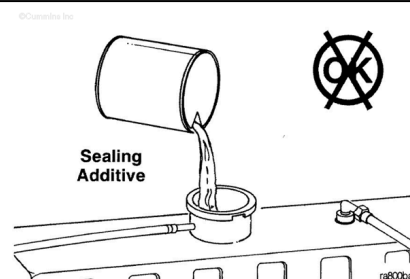
Do **not** use a floating ball hydrometer. Floating ball hydrometers can give incorrect readings.



Cooling System Sealing Additives

Do **not** use sealing additives in the cooling system. The use of sealing additives will:

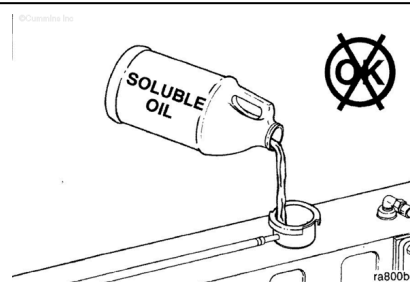
- Build up in coolant low flow areas
- Clog coolant filters
- Plug radiator and oil cooler.



Cooling System Soluble Oils

Do **not** use soluble oils in the cooling system. The use of soluble oils will:

- Allow cylinder liner pitting
- Corrode brass and copper
- Damage heat transfer surfaces
- Damage seals and hoses.



Fleetguard® DCA4 Service Filters and Liquid Precharge

DCA4 Service Filters		DCA (Fleetcool) Service Filters	
Part Number	SCA Units	Part Number	SCA Units
WF2070	2	WF2050	2
WF2071	4	WF2051	4
WF2072	6	WF2052	6

DCA4 Service Filters		DCA (Fleetcool) Service Filters	
WF2073	8	WF2053	8
WF2074	12	Not available	12
WF2075	15	WF2054	15
WF2076	23	WF2055	23
WF2077	(blank filter without SCAs)	WF2077	(blank filter without SCAs)

DCA4 Liquid	DCA (Fleetcool) Liquid				
Part Number	Size	SCA Units	Part Number	Size	SCA Units
DCA60L	0.47 liter [1 pt]	5	DCA30L	0.47 liter [1 pt]	5
DCA65L	1.89 liter [2 qt]	20	DCA35L	1.89 liter [2 qt]	20
DCA70L	3.78 liter [1 gal]	40	DCA40L	3.78 liter [1 gal]	40
DCA75L	18.9 liter [5 gal]	200	DCA45L	18.9 liter [5 gal]	200
DCA80L	208 liter [55 gal]	2200	DCA50L	208 liter [55 gal]	2200

Install service filter(s) and/or liquid containing the number of SCA units below:

Maintenance Intervals for Cooling Systems up to 76 liters [20 gal]						
Service Interval			System Size in liters [gal]			
Kilometers	Miles	Hours	4-19 [1-5]	19-38 [6-10]	42-57 [11-15]	60-76 [16-20]

System Size in liters [gal]										
Hours	79-144 [21-30]	117-189 [31-50]	193-284 [51-75]	288-378 [76-100]	382-568 [101-150]	572-757 [151-200]	761-946 [201-250]	950-1135 [251-300]	1139-1325 [301-350]	1329-1574 [351-400]
751-1000	25	50	80	100	150	200	250	300	350	400
501-750	20	35	60	75	110	150	190	225	260	300
251-500	15	25	40	50	75	100	125	150	175	200
0-250	10	15	20	25	40	50	65	75	90	100

Notes:

- A. Consult the vehicle equipment manufacturer's maintenance information for total cooling system capacity.
- B. When draining and replacing the coolant, **always** pre-charge the cooling system to a SCA level of 1.5 units per gallon. This concentration level **must not** be allowed to go below 1.2 units and **must** be controlled when the level is greater than 3 units. Action needed when the level goes below 1.2 is a filter and liquid pre-charge; from 1.2 to 3.0 units, filter **only**; above 3.0, test at every oil change until level falls to 3.0 or below.

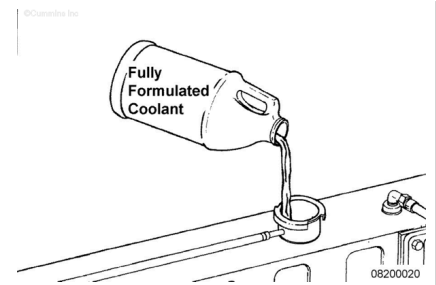
When performing service which requires draining the cooling system, take special precautions to collect it in a clean container, seal it to prevent contamination, and save for reuse.

- A. Change coolant filters at each oil change to protect the cooling system. Consult the coolant capacity chart to determine the correct coolant filter for a given cooling system capacity and oil drain interval.

Supplemental Coolant Additive (SCA)

Fully formulated products contain SCA's and are required to protect the cooling system from fouling, solder blooming, and general corrosion. The coolant filter is required to protect the

coolant system from abrasive materials, debris, and precipitated coolant additives.



Supplemental coolant additives, or equivalent, are used to prevent liner pitting, corrosion, and scale deposits in the cooling system.

Use the correct Fleetguard® coolant filter to maintain the recommended SCA concentration in the system.

Maintain the correct concentration by changing the service filter at each oil drain interval.

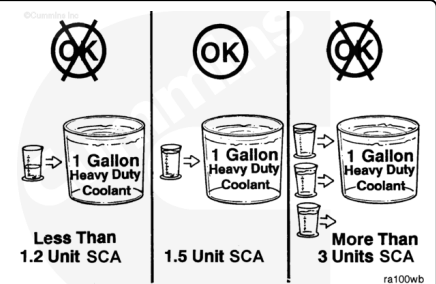
The correct filter is determined by the total cooling system capacity and oil drain interval. Refer to the Coolant Capacity Charts.



⚠ CAUTION ⚠

Insufficient concentration of the coolant additives will result in liner pitting and engine failure.

The SCA concentration **must not** fall below 1.2 units or exceed 3 units per gallon of cooling system capacity.



Use the correct Fleetguard® coolant filter to maintain the recommended SCA concentration in the system.

Maintain the correct concentration by changing the service coolant filter at each oil drain interval.

Note : The correct filter is determined by the total cooling system capacity and oil drain interval.



Testing SCA Concentration Level CC-2602 Test Kit

Carefully follow the instructions to test the coolant and take the appropriate action recommended by the kit.

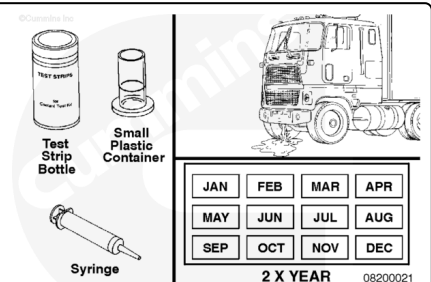
Precautions and Instructions for Proper Kit Use

- The coolant sample to be tested **must** be between 10 and 54°C [50 and 130°F]. If the sample is too cold or too hot, you will get incorrect results.
- To get the best color match results, compare test strip pads to the color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test does fall between two colors on the color chart, choose the lower numbered block. It is safer to underestimate your results than to overestimate.
- The test strips do have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect the life of the strips.
- Keep the cap tightly sealed on the test strip bottle except when removing a strip. Store away from direct sunlight and in an area where the temperature will generally stay below 32°C [90°F].
- Do **not** use the test strips after the expiration date stamped on the bottle.
- Discard the kit if any of the pads on the unused strips have turned light brown or pink.
- Use one strip at a time and take care **not** to touch any of the pads on the strip. Doing so will contaminate the pads and affect the test results.
- If the strip container is left uncapped for 24 hours, moisture in the air will render the strips useless, although no discoloration will be evident.
- **Only** use the color chart supplied with the kit.
- Clean and dry the sample cup and syringe after each use. This will prevent contaminating future samples.
- Following the correct test times is very important. Use a clock or stopwatch.

Test Intervals

Testing is recommended if the operator is **not** sure of his cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

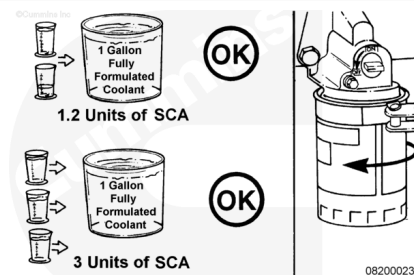
Testing is also recommended twice a year to monitor the SCA level. If the SCA level is above 3 units, test at subsequent oil drain intervals until the concentration is back under 3 units. When the concentration is back under 3 units, start installing the correct service filters at each drain interval.



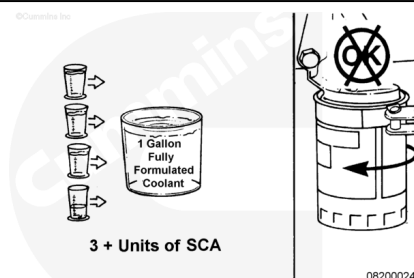
If the concentration is below 1.2 units per gallon, replace the filter and precharge with liquid.



If the concentration is 1.2 to 3 units per gallon, replace the filter.

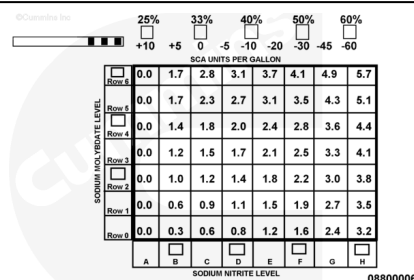


If the concentration is above 3 units per gallon, do **not** replace the service filter. Test the coolant at subsequent oil drain intervals until the concentration is back under 3 units. When the concentration is back under 3 units, start installing service filters at each oil change interval.



Do **not** utilize the test kit to maintain minimum SCA concentration levels (i.e., 1.5 units).

In some instances the A or B reading can be high. However, it is the combined reading that is important. Therefore, always follow the chart.



Cummins Inc

Cummins: 1-800-DIESELS
1-800-521-4005

1-800-22-FILTERS
1 - 800 - 223 - 4583

00200003

Call the numbers listed above to get answers to any questions you have about cooling system maintenance.

CC2602 Coolant Test Kit:

- Works with any SCA formulation (Call 1-800-521-4005 if you have this test kit and the color chart does **not** show the number of units of SCA per gallon of coolant).

Probablizer:

- 3318169S Plug - Installs on the engine for easy coolant sampling.
- 3318168S Cap - Use with Monitor C bottle to sample coolant.
- CC2700 Monitor C - Lab analysis of coolant samples.

Coolant Replacement Requirements

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain and flush the cooling system after 6,000 hours or 2 years of service. Refill with either new fully formulated coolant or a 50/50 mixture of high quality water and fully formulated antifreeze, and install the correct service coolant filter.

